

Arc Welding Controller

G4 Controller Series

Further evolved welding functions and improved compatibility with peripheral devices



Panasonic
GREEN
IMPACT

1 Further evolved welding performance

- 261 types of welding tables included (1.7 times the conventional models)

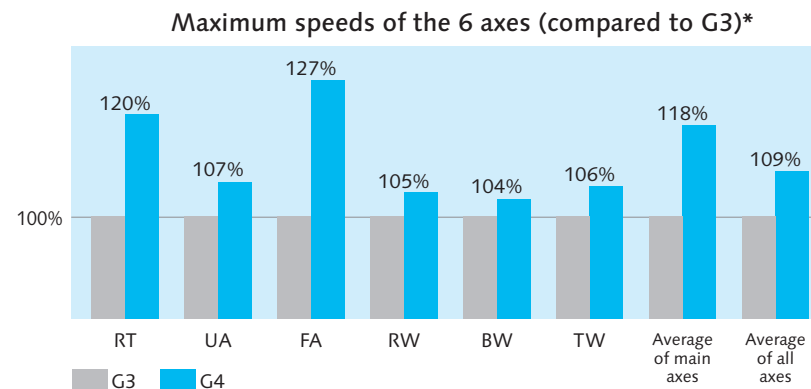


Mild steel: 95 types
Stainless steel: 42 types
Stainless steel (ferrite-based): 34 types
Hard aluminum: 31 types
Zinc-plated steel: 26 types
Soft aluminum: 18 types

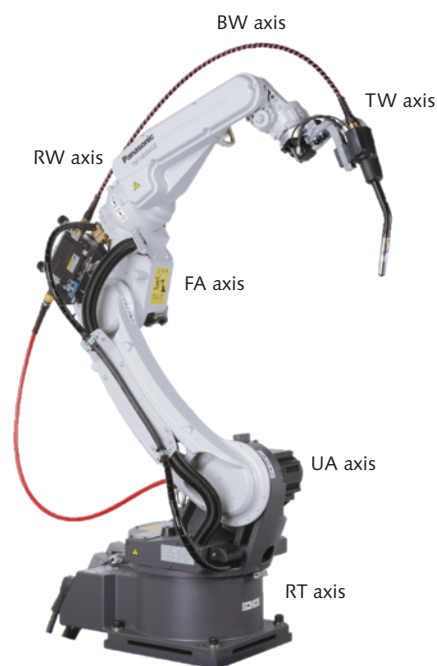
*The above list represents a portion of the types.
*Tables will be added as necessary.
The number of tables include optional ones.

2 Optimized operation reduces the time required to move to the next weld point

- The maximum speed of each axis has been improved by up to 27% (compared to the G3 controller)
- The basic performance has been enhanced through improved CPU performance and memory capacity
- The maximum speeds of all axes have been enhanced through improved acceleration and deceleration control

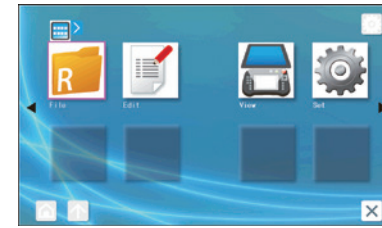


*The above are the TM-1400 test results (under our test environment).

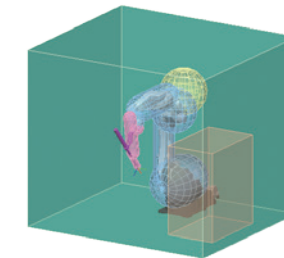


3 Touch interactions and 3D display improve ease of use

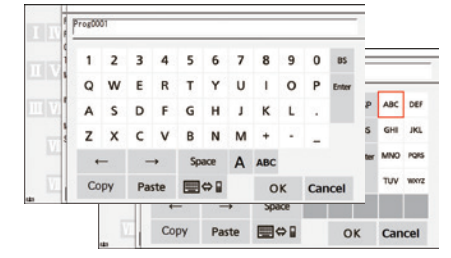
- The touch panel is operable while wearing gloves
- 3D engine allows finer 3D display and intuitive operation
- Character enlargement function improves visibility



New teach pendant screen with a touch panel operable while wearing work gloves



Fine 3D display on LCD with a resolution 1.6 times the conventional model

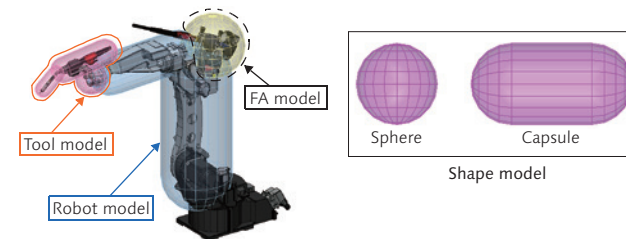


Intuitive operation simplifies text entry

4 Software-based safety mechanism enables more flexible and safer work environments

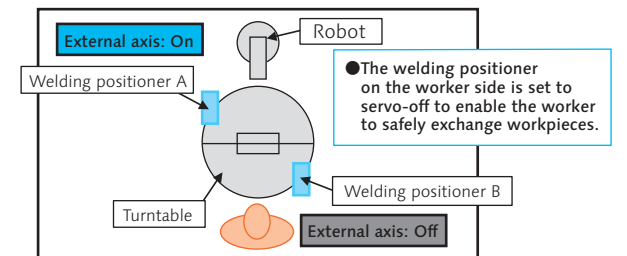
Area monitoring function

Monitors whether the spherical or capsule-shaped models arranged on the manipulator and tool are within the safety area. When the shape models are outside the specified safety area, an error is triggered to alert operators of unsafe conditions and halt the robot operation.



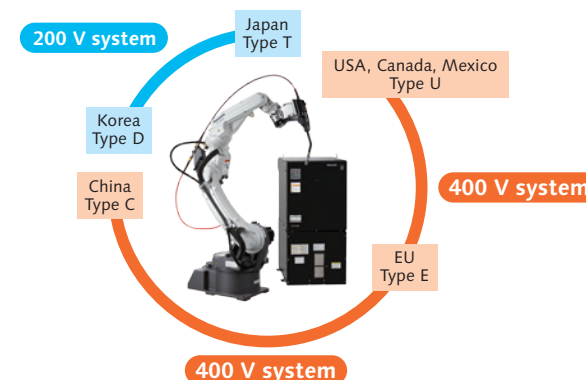
Individual servo-off function

The individual servo-on/off function for external axes enhances the safety of workers. In the example below, two welding positioners are on the turntable. The operation of welding positioner A, where the robot is welding, is on. At that time, welding positioner B is turned off to allow the worker to safely exchange workpieces.

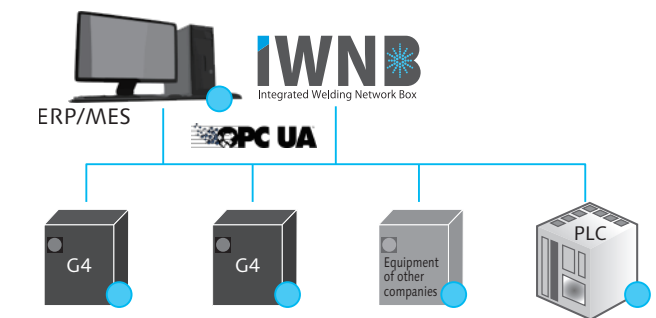


5 The 400 V systems (380 to 460 V) as well as the 200 V systems (200/220 V) are available

- No step-down transformer is required, even in factories with different input voltages



6 The conformance to the OPC UA standard facilitates integration with peripheral devices



Please refer to the website for other ancillary devices and details.



Achieves high-quality welding

TS Series

Space saving & high payload

TS
800
950

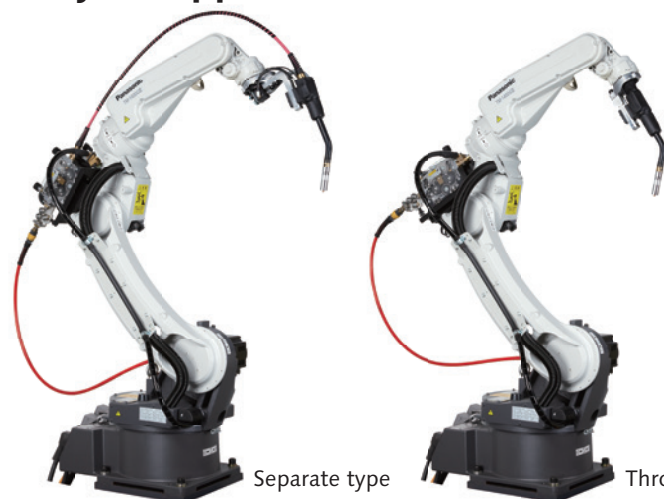


External type/
Through-arm type

TM Series

The torch type can be selected to suit your application

TM
1100
1400
1600
1800
2000



Separate type

Through-arm type

TL Series

Long arm & high payload

TL
1800
2000



External type

LA-1800

A single robot can perform material handling and welding operations

LA
1800



External type

Manipulator lineup

	TS Series		TM Series					TL Series		LA
	800	950	1100	1400	1600	1800	2000	1800	2000	1800
Separate	—	—	○	○	○	○	○	—	—	—
Through-arm	○	○	○	○	○	○	○	—	—	—
External	○	○	*1	*1	—	—	—	○	○	○
Payload	8 kg		6 kg		4 kg	6 kg		8 kg	6 kg	26 kg

* Please contact us for products that comply with C-UL, UL, CE, KCS, and CCC standards.

*1 Supported for TIG and some other types

Various features specialized for arc welding

1 Enhanced basic performance

Increased motion speed (reduced takt time)

The maximum speed of each axis has been improved by up to 27% (compared to the G3 controller)

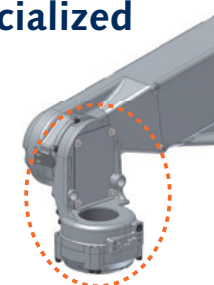
Extended maximum reach (applicable welding range)

TM-1400: 1 437 mm (63 mm more than the conventional TA type)

2 Arm structure specialized for welding

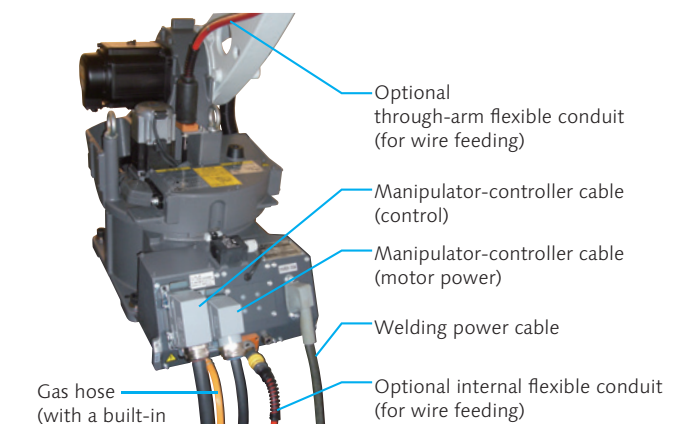
Side mount arm structure

Makes the arm compact and improves accessibility to workpieces



3 Structure designed specifically for welding

Tidy appearance with through-arm cables



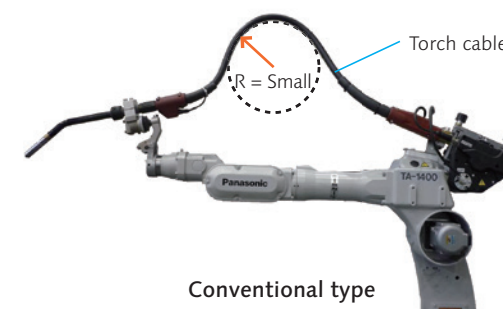
* The optional internal flexible conduit is for use with a pail-pack wire only.

Separate type (TM Series)

The advantages of both the through-arm type and external type torch cables are achieved in a well-balanced manner.

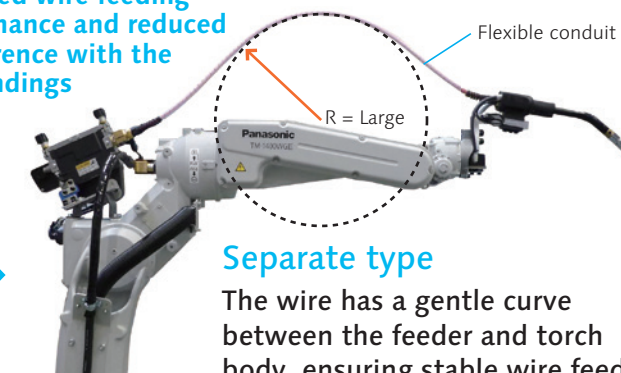


1 External flexible conduit



Conventional type

Improved wire feeding performance and reduced interference with the surroundings



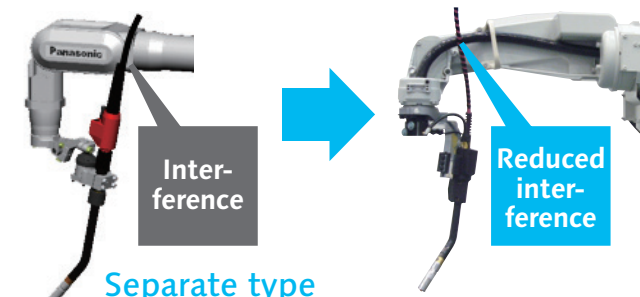
Separate type

The wire has a gentle curve between the feeder and torch body, ensuring stable wire feed.

2 Through-arm power cable

Conventional type

Power cable may interfere with the surroundings depending on the welding position.

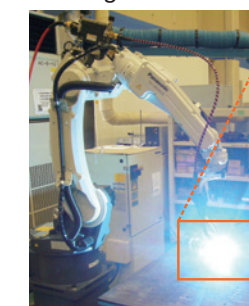


Separate type

Power cable is built into the manipulator to reduce interference with the surroundings.

Separate type: Example of circumferential welding

Suppresses wire twisting.



Reduces wire target position misalignment at the weld start and end points.

New welding robot configuration offers even higher quality welding.

8

Seamlessly synchronizes 3 elements: welding output, wire feed speed, and weaving movement.

Alternates between conditions A and B during spiral weaving (ideal for welding plates of different thicknesses (high current for a thick plate, low current for a thin plate)).

TAWERS[®]

WG4/WGH4

TAWERS enables flexible welding process selection/switching

SP-MAG II for MAG welding
(short-circuit transfer range for thin plates)
MTS-CO₂ for CO₂ welding

SP-MAG II

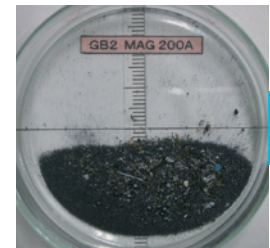
SP(Super-imposition) Control

Reduces spatter significantly during MAG welding of thin plates

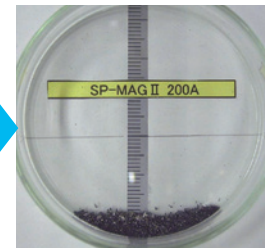
Welding waveform control technology achieves low spatter in short-circuit transfer range.

■ Spatter generation comparison (1 minute at 200 A)

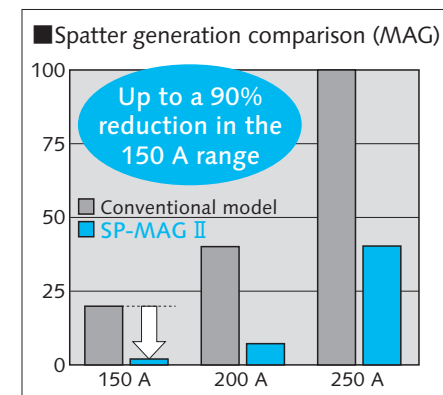
Full Digital Welding Machine
(Conventional model)



TAWERS (SP-MAG II)



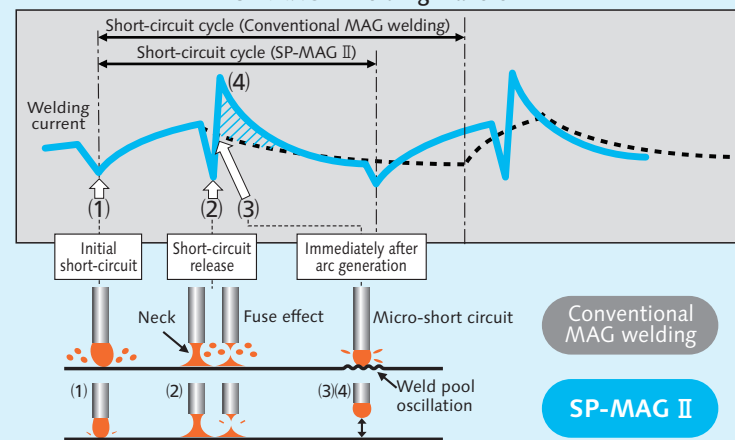
Spatter reduction effect



Wire used: YM-50MT



SP-MAG II welding waveform



(1) Initial short-circuit control

Detects an initial short-circuit accurately and then enables secondary switching^{*1} to rapidly reduce the welding current to prevent a micro-short circuit that causes spatter, and ensure short circuiting transfer.

(2) Neck control

Detects a neck of the wire tip and then enables secondary switching^{*1} to rapidly reduce the welding current to prevent the fuse effect of the wire tip that causes spatter.

(3) HS control

Suppresses the weld pool oscillation immediately after arc generation, and prevents a micro-short circuit that causes spatter.

(4) SP control

Superimposes the current immediately after short-circuit release to increase the melting rate of the wire tip, thereby making the next short-circuit smoother and shortening the cycle.

^{*1} Secondary switching

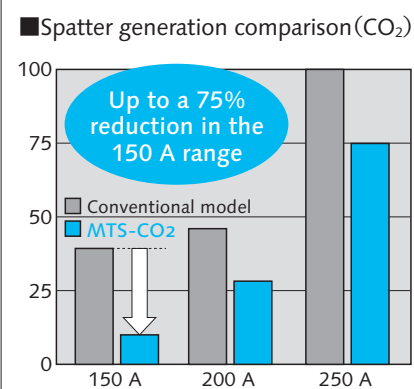
Spatter reduction process that rapidly reduces welding current immediately before and after a short-circuit, and enables a smooth transition between the arc and short circuit.

MTS-CO₂

MTS(Metal Transfer Stabilization) Control

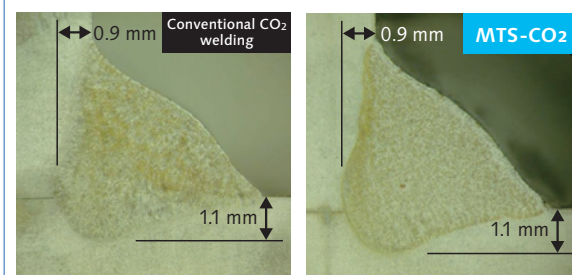
Reduces spatter by up to 75% using CO₂ gas

MTS control added to our SP-MAG technologies reduces spatter generation specific to CO₂ welding.



Stable pan-bottom shaped penetration achieved by CO₂ welding.

Penetration comparison example



Joint: Fillet Base metal: Mild steel SPCC (2.3-mm thick, 120 A)
Welding speed: 30 cm/min
Wire: YGW12 (1.2 mm) Gas: CO₂

Conventional CO₂ welding



MTS-CO₂



TAWERS[®]

WG4/WGH4

TAWERS enables flexible welding process selection/switching

Pulse MAG welding (high-current range)
HD-Pulse for high-speed and low-spatter welding
Normal-Pulse for low-spatter welding of medium and thick plates

HD-Pulse

HD-Pulse(Hyper Dip-Pulse Control)

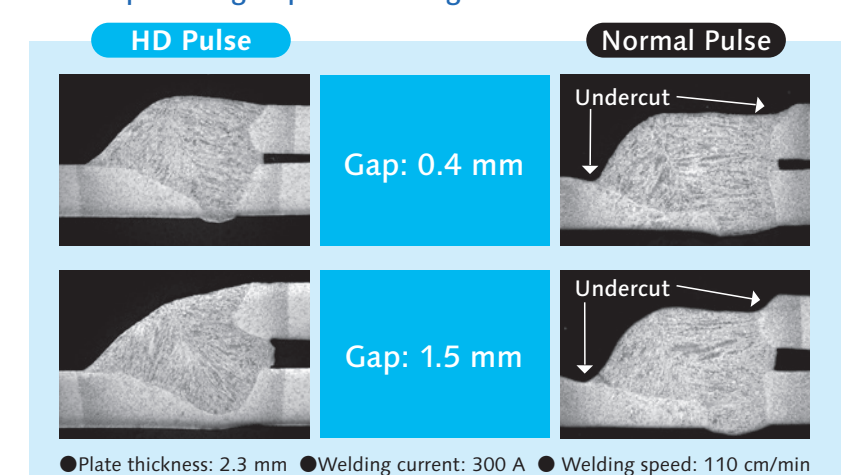
Achieves high-speed pulse welding

Short arc length and narrow arc width prevents undercuts caused by insufficient deposition during high-speed welding.

HD-Pulse welding features

- Prevents undercuts during high-speed welding.
- The short-circuit transfer enables lower heat input than drop transfer. Gap tolerance is improved.
- Precisely controls dip timing, reducing spatter.

Example of high-speed welding



Preventing undercuts with ideal penetration

Types of droplet transfer

HD-Pulse control

Transfer type: 1 dip by 1 pulse
(short-circuit transfer)

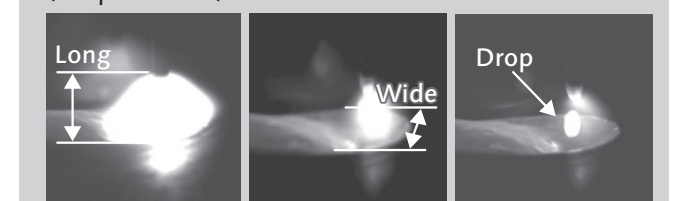


Process comparison in spray transfer range (280 A or more)

Welding process	SP-MAG II	Normal-Pulse	HD-Pulse
Welding speed	Good	Good	Excellent
Spatter	Average	Excellent	Good
Penetration pattern	Marginal	Average	Excellent
Undercut	Marginal	Marginal	Excellent
Base metal heat input	Marginal	Marginal	Good
Gap handling	Marginal	Marginal	Good
Overall evaluation	Marginal	Marginal	Excellent

Normal-Pulse control

Transfer type: 1 drop by 1 pulse
(drop transfer)



● SP-MAG II:

Spatter control is a challenge in the high-current range.

● Normal-Pulse :

Undercut control is a challenge in high-speed welding.

HD-Pulse control is ideal for high-current and high-speed welding